Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahaj-bhadja sahaj-bhadja

Summary

Software engineer with 2+ years building AI-powered products, TypeScript/React interfaces, and backend APIs. Strong in shipping LLM-enabled workflows, SQL-backed systems, and polished full-stack features across fast-moving product environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React interfaces and Flask APIs for an AI grading product handling 10,000+ daily requests, integrating uploads, grading history, and secure file-processing workflows across fast-moving product requirements.
- Integrated multimodal LLMs, OCR, and concurrent batch-processing into production workflows, wiring frontier models into instructor-facing product flows that reduced feedback turnaround by 60% while keeping backend behavior stable under peak load.
- Tracked model usage, request costs, and grading outputs across web workflows, helping teams make product tradeoffs quickly while iterating on real automation features instead of isolated research prototypes.

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Maintained SQL-backed platform services for 500+ users, tuning indexed queries, background workers, and validation flows across FastAPI workflows to improve dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant async AI orchestration and data-integration workflows across application, database, and storage layers, supporting 1K+ AI jobs and recurring releases while improving production reliability in fast-changing environments.
- Translated stakeholder requests into maintainable backend changes, documenting debugging paths and release fixes that made recurring production issues faster to isolate, resolve, and ship without disrupting active platform operations.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered API and ETL integrations using Python, SQL, AWS Glue, and S3, improving response time by 30% while supporting scalable service integrations, data workflows, and maintainable backend components.
- Maintained 20+ automated tests and CI pipelines with Jenkins, PyTest, and Mockito, increasing release reliability by 68% while strengthening debugging discipline, deployment consistency, and collaboration across distributed engineering environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built MERN platform features with REST APIs, WebSockets, and responsive interfaces, supporting synchronized records across modular workflows while translating business requirements into reliable end-to-end production application functionality.
- Automated Docker-based build and test flows with Jenkins, cutting release time by 50% and improving deployment consistency, troubleshooting discipline, and shared development practices across distributed product environments.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C++, Bash
Frontend: React, Next.js, HTML, CSS, TailwindCSS, responsive design, UI state
Backend & APIs: Node.js, Express, Flask, FastAPI, REST APIs, WebSockets, PostgreSQL, MySQL, MongoDB
AI & Automation: LLM integration, prompt workflows, OCR, multimodal models, async workers, model tracking
Cloud & Delivery: AWS, Azure, Docker, Jenkins, GitHub Actions, CI/CD, SQL Server, Azure SQL
Engineering: object-oriented programming, unit testing, debugging, code reviews, Agile/Scrum, system design, scalability

Projects

Chat2SQL Code | *Java, TypeScript, SQL, PostgreSQL, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, combining Java and TypeScript services to accelerate query authoring, schema exploration, and structured database management from a unified interface.

PiXMod Live | *Next.js, MongoDB, Stripe, CLIP, SAM*
- Built an AI-powered SaaS image platform with Next.js, MongoDB, Stripe, and vision models, supporting premium workflows, scalable asset storage, and polished interfaces while increasing premium conversion by 18%.

HMIS Live | *MERN, TypeScript, ABDM APIs, Twilio, Sentry*
- Built a healthcare workflow platform with TypeScript, MERN, ABDM APIs, Twilio alerts, and Sentry monitoring, reducing administrative workload by 30% through responsive scheduling, consent-aware access, and operational dashboards.